Program: B Tech and MBA Tech Data Science				**Semester :** V	
Course: Applied Time Series Analysis				**Code :**	
Teaching Scheme				**Evaluation Scheme**	
Lecture (Hours per week)	**Practical (Hours per week)**	**Tutorial (Hours per week)**	**Credit**	**Internal Continuous Assessment (ICA) (Marks -50)**	**Term End Examinations (TEE) (Marks - 100)**
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50

Pre-requisite: Statistical Structures in Data and Inference

Course Objective

This course provides an introduction to time series analysis. It will educate students on how to choose an appropriate forecasting method in a particular environment and then apply it using necessary computations. It aims to equip students with various forecasting techniques and knowledge on modern statistical methods for analyzing time series data. It will also provide directions to Improve forecasting using better statistical models based on statistical analysis

Course Outcomes

After completion of the course, students will be able to –

1. Explain the fundamental components of time series data, such as trend, seasonality, and autocorrelation, and visualize them using various plotting techniques.
2. Apply time series regression models, exponential smoothing, ARMA, ARIMA, and SARIMA models to forecast univariate and seasonal data.
3. Analyze stationary and non-stationary time series processes using ACF, PACF, unit root tests, and diagnostics to identify appropriate forecasting models.
4. Evaluate forecasting performance of hierarchical and grouped time series models using reconciliation methods and compare with alternative approaches.
5. Design advanced forecasting solutions using complex seasonality, the Prophet model, VARs, bootstrapping, and bagging on real-world datasets using statistical software.

Detailed Syllabus

Unit	Description	Duration
1.	**Examples of Time Series** The basic steps in a forecasting task **Time Series Graph** Seasonal plots, Seasonal subseries plots, Lag plots, Autocorrelation and time series decomposition, Stationary models and autocorrelation function Estimation and elimination of trend and seasonal components	**03**
2.	**Time Series Regression Models and Exponential Smoothing**	**02**
3.	**Stationary Process and ARMA Models** Basic properties and linear processes, Introduction to ARMA models, properties of sample mean and autocorrelation function, Forecasting stationary time series ARMA(p, q) processes, ACF and PACF, Forecasting of ARMA processes.	**05**
4.	**Modeling and Forecasting with ARMA Processes** Preliminary estimation, Maximum likelihood estimation, Diagnostics, Forecasting Order selection.	**05**

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5.	**Nonstationary and Seasonal Time Series Models** ARIMA models, Identification techniques, Unit roots in time series, Forecasting ARIMA models, Seasonal ARIMA models, Regression with ARMA errors	**03**
6.	**SARIMA - Seasonal Autoregressive Integrated Moving Average**	**02**
7.	**Forecasting Hierarchical and Grouped Time Series** Hierarchical and grouped time series, Single level approaches, Forecast reconciliation	**03**
8.	**Advanced Forecasting Methods** Complex seasonality, Prophet model, Vector auto regressions, Bootstrapping and bagging	**07**
	Total	**30**

Text Books

1. Galit Shmueli and Kenneth C. Lichtendahl , *Practical Time Series Forecasting with R: A Hands-On Guide*, 2nd Edition, Axelrod Schnall Publishers, 2016.
2. Damodar Gujarati, Dawn Porter, Sangeetha Gunasekar, Basic Econometrics, 5th Edition, McGraw Hill Education, 2017.

Reference Books

1. Jenkins and Reinsel, *Time Series Analysis: Forecasting and Control*, 5th Edition, Prentice Hall, 2015.
2. Robert Shumway, David Stoffer, *Time Series Analysis and Its Applications with R examples*, 4th Edition, Springer, 2016.
3. Rami Krispin , *Hands-On Time Series Analysis with R: Perform time series analysis and forecasting using R*, 1st Edition, Packt Publishing Limited, 2019.

Laboratory Work

8 to 10 programming exercises (and a practicum) based on the syllabus.

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